

Story2_DATA608

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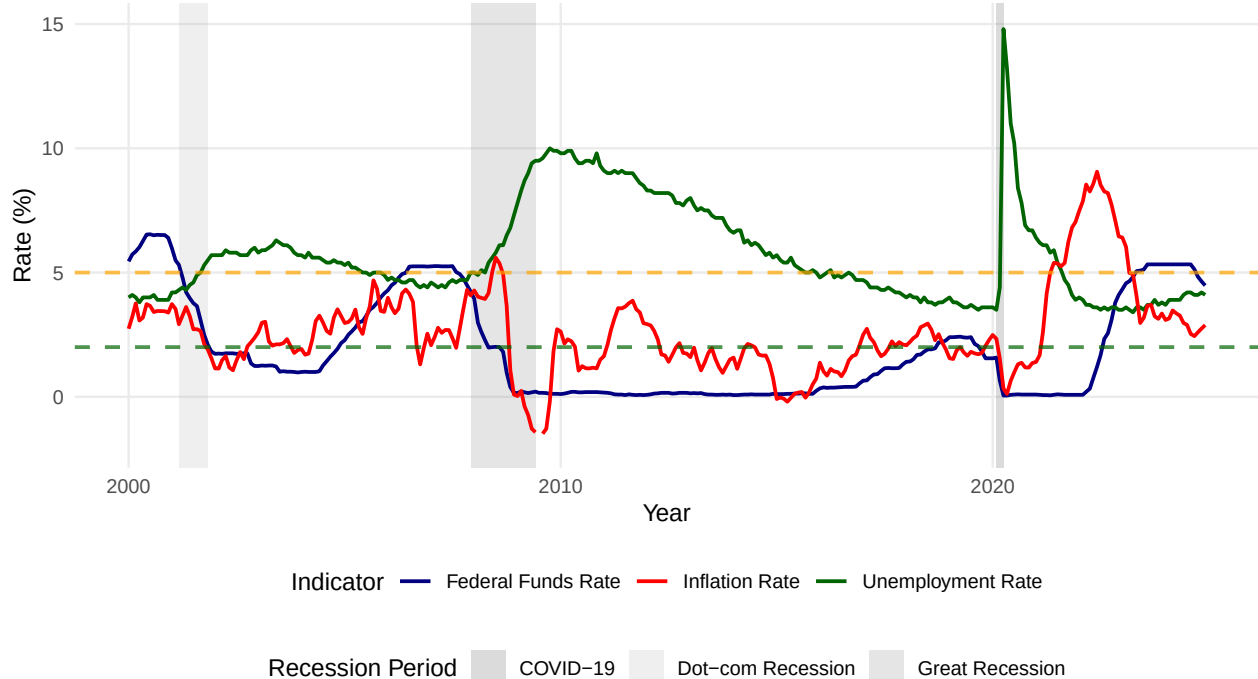
INTRODUCTION

The Federal Reserve operates under a dual mandate from Congress: **maintain price stability** (control inflation) and **promote maximum employment** (maintain low unemployment). These objectives often appear contradictory—policies that cool inflation (raising interest rates) can slow economic growth and increase unemployment, while policies that boost employment (lowering rates) can fuel inflation.

This assignment analyzes 25 years of economic data (1999-2024) to answer a critical question: “Has the Fed been able to fulfill the mandate given to it by Congress?”

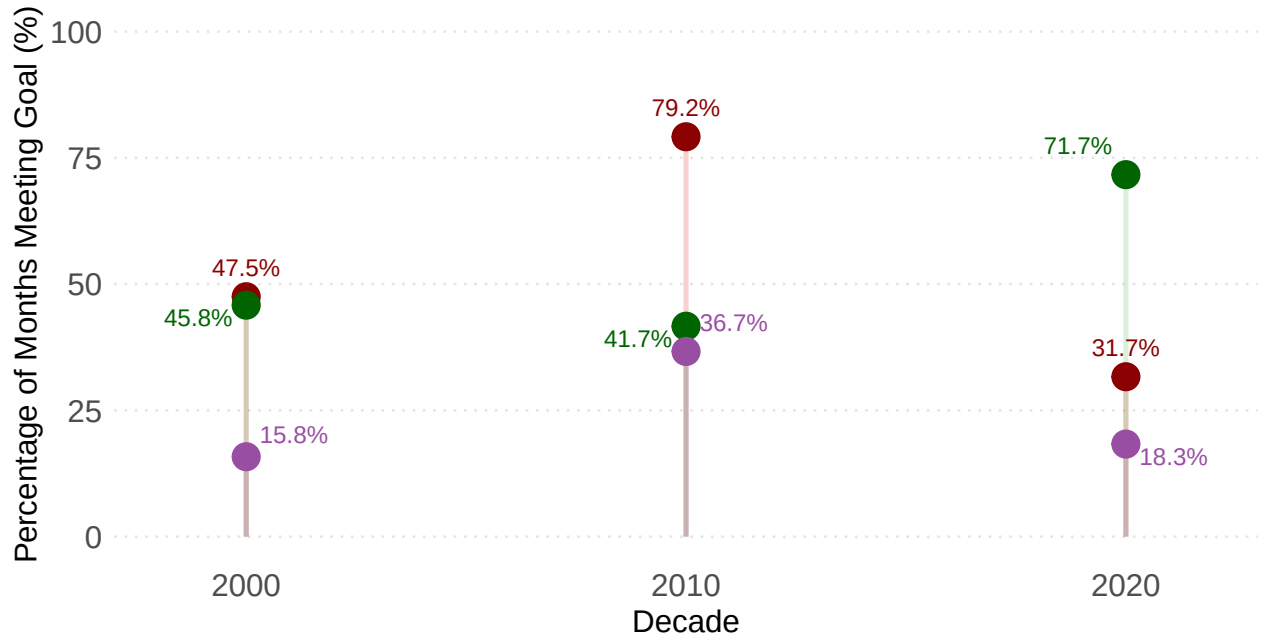
The Fed's Dual Mandate: 25 Years of Balancing Inflation and Unemployment

Shaded areas indicate recessions | Dashed lines show target thresholds



Fed Mandate Performance by Decade

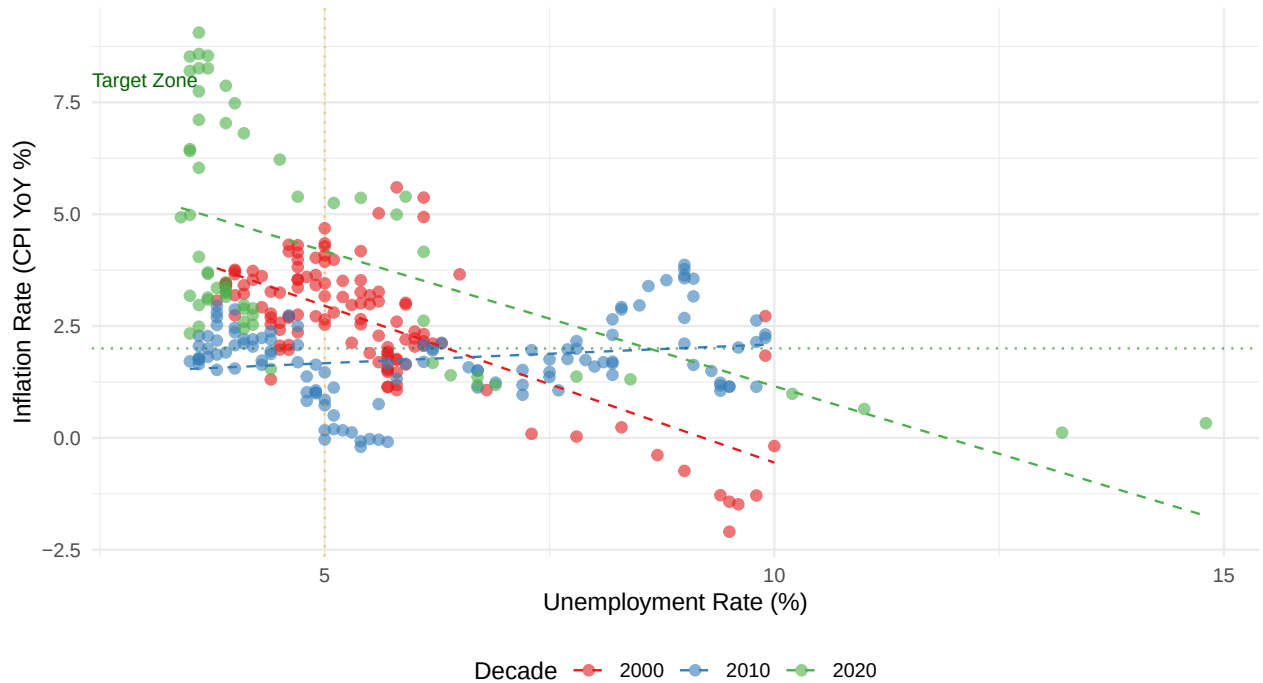
Height shows % of months meeting each goal



Performance Metric ● Both Goals ● Inflation Goal ● Unemployment Goal

The Inflation–Unemployment Trade-off (Phillips Curve)

Each point represents one month (1999–2024)



##

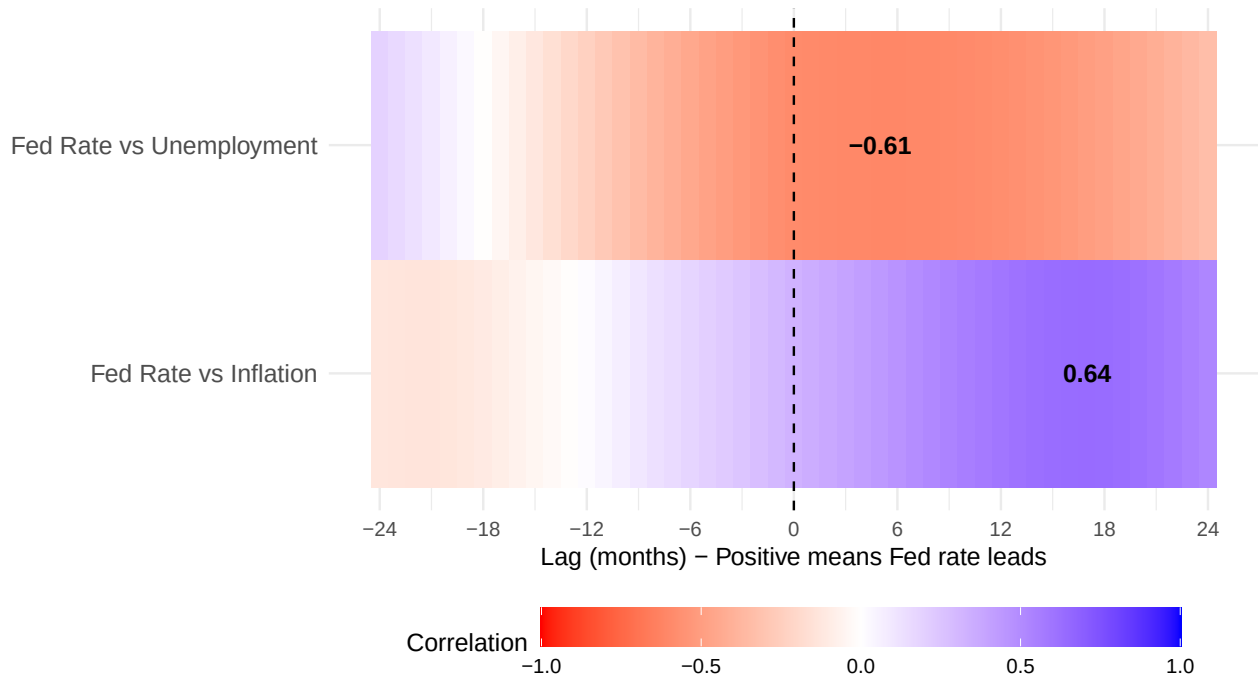
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FED MANDATE PERFORMANCE SUMMARY (1999–2024)

```
##
## =====
##   Total Months Avg Inflation Avg Unemployment Avg Fed Funds Rate
## 1           300         2.58%           5.69%           1.92%
##   Inflation Goal Met Unemployment Goal Met Both Goals Met Neither Goal Met
## 1           57.0%           49.3%           24.7%           18.3%
##
## =====
## DECADE-BY-DECADE BREAKDOWN
##
## =====
## # A tibble: 3 x 6
##   decade `Avg Inflation` `Avg Unemployment` `Inflation Goal` `Unemployment Goal`
##   <dbl> <chr>           <chr>           <chr>           <chr>
## 1    2000 2.57%           5.54%           47.5%           45.8%
## 2    2010 1.77%           6.22%           79.2%           41.7%
## 3    2020 4.21%           4.95%           31.7%           71.7%
## # i 1 more variable: `Both Goals` <chr>
##
## All visualizations created successfully!
```

When Do Fed Actions Show Their Effect?

Cross-correlation at different time lags | Positive lag = Fed rate change comes first

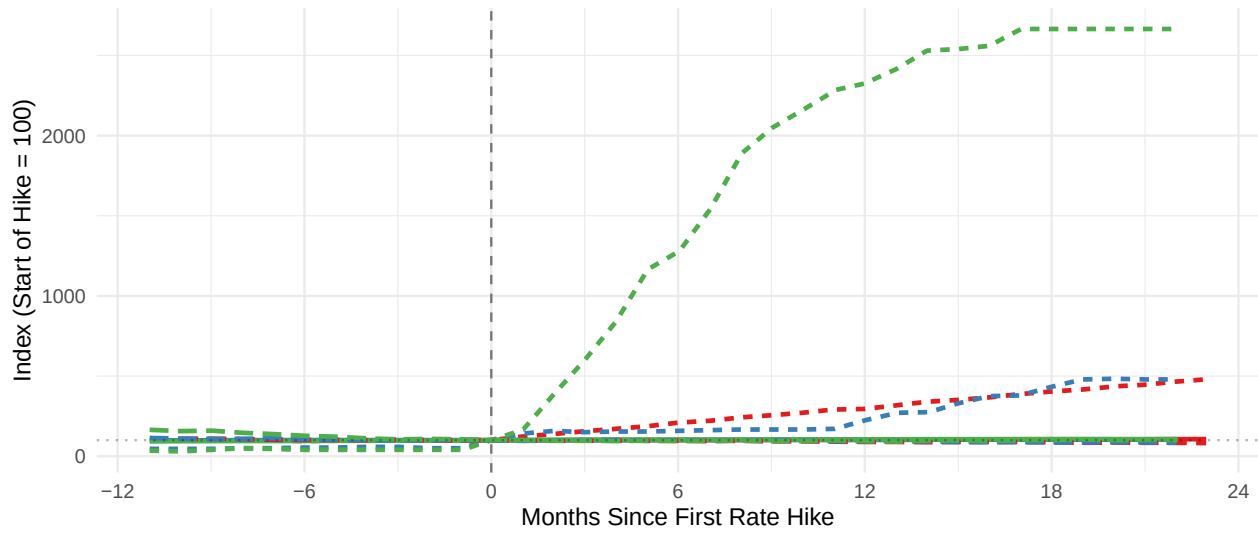


Interpretation: Positive lag means Fed rate change precedes the effect

```
## Processing: 2004-2006 Tightening
## Processing: 2015-2018 Normalization
## Processing: 2022-2023 Inflation Fight
```

What Happens After the Fed Starts Raising Rates?

Indexed to 100 at start of hiking cycle

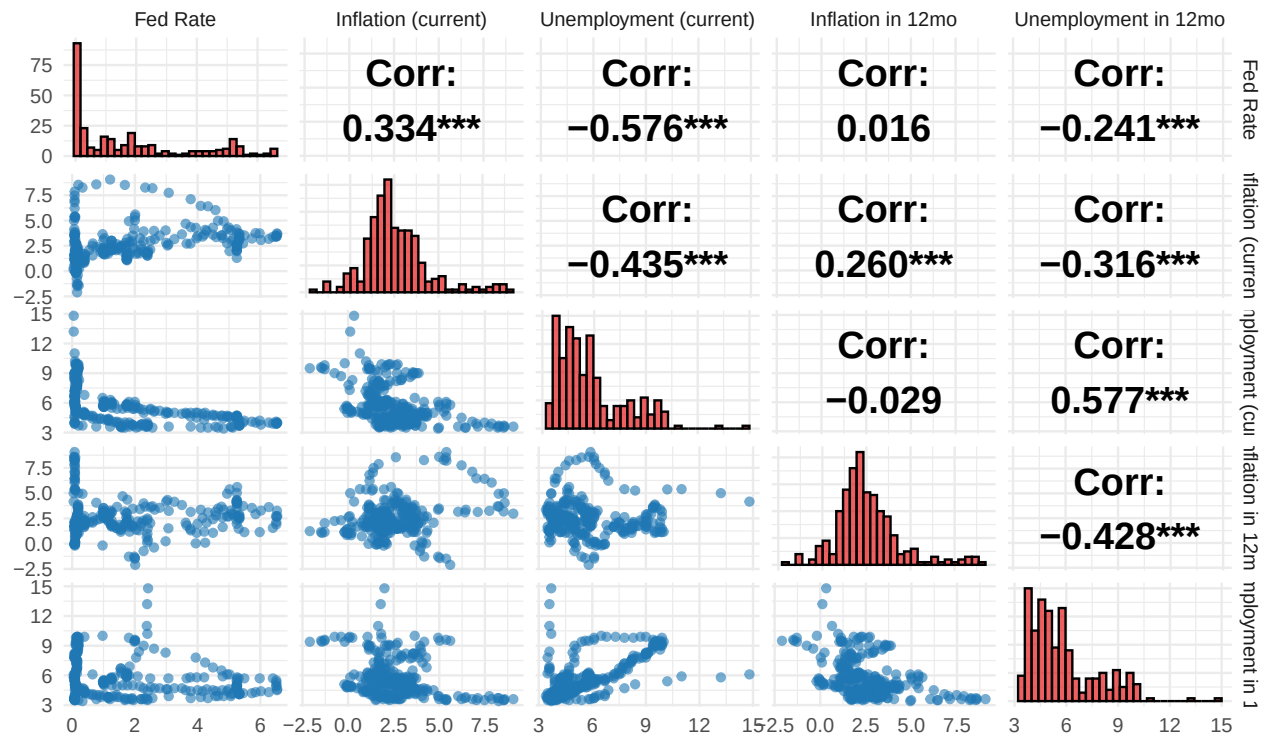


Hiking Episode — 2004–2006 Tightening — 2015–2018 Normalization — 2022–2023 Inflation Fight

Indicator — CPI (Inflation) — Fed Funds Rate — Unemployment

The Ripple Effect: How Fed Policy Propagates Through the Economy

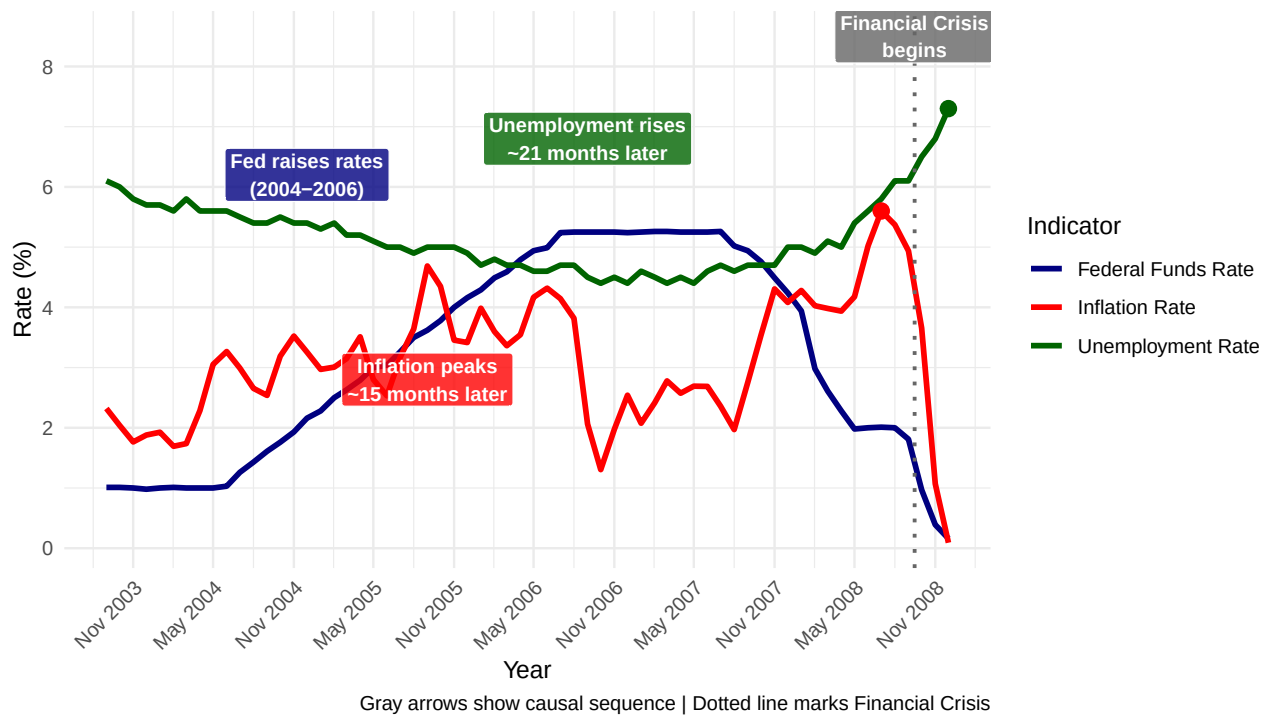
Lead-Lag Relationships: Fed Actions and Economic Outcomes



The Fed's casual chain

The Fed's Causal Chain: Rate Hikes → Lower Inflation → Higher Unemployment

2004–2008 cycle shows typical time lags | Inflation peaks before unemployment rises



Conclusion:

The Fed has been a capable crisis manager but an inconsistent mandate fulfiller. The dual mandate remains an aspirational target—achievable only about a quarter of the time—rather than a consistently attainable goal.

Recommendations for Policymakers

- Communicate the lag effect more clearly to the public
- Set realistic expectations that both goals cannot always be met
- Consider a flexible timeline for achieving both mandates rather than simultaneous success

“Monetary policy is a balancing act—and the Fed walks a tightrope between inflation and unemployment every single day.”